

International Cybersecurity and Digital Forensics Academy (ICDFA)

Course Code: GRC102 – Information Security Governance

Week 4 Practical Laboratory

Linux Security Monitoring and Auditing

Audit and Control Assurance Report

Student Name: Mercy Ajayi

Registration Number: C11/26/CGRCE/17219

Date of Submission: October 2, 2026

1. Executive Summary

This report covers the Week 4 practical lab on Linux security monitoring and auditing. I completed the lab in my assigned Ubuntu Practice Hub workspace (hostname: student000272) on 2 October 2026. In the scenario, I worked as a Security Control Assurance Analyst, checking whether the monitoring controls on the Linux system were actually working and documenting the evidence I found.

Overall posture. The monitoring controls on the system are weak. Although the required tools are installed, most of the monitoring is not functioning. `auditd` is installed and I created audit rules, but the service would not start and the kernel refused to load the rules. This means there is no audit evidence available. The system journal is also empty, the usual authentication and system log files are missing, and the login record files are empty. Lynis reported a hardening index of 62 out of 100, with 3 warnings and 43 suggestions.

One important observation. The workspace appears to be running as a container rather than a full virtual machine. PID 1 is `sleep` instead of `systemd`. This may explain why `auditd` and other logging services are not working, but I could not confirm the cause from inside the workspace. I therefore treat it as a likely explanation rather than a confirmed fact. The findings in this report are based on the evidence I was able to observe directly.

Top three findings

1. **W4-F01 (High): Audit logging is configured but not functioning.** `auditd` is installed and the rules file exists, but the service fails to start, the kernel refuses the rules, and no audit log has been created.
2. **W4-F02 (High): System and authentication logs are not being collected.** The journal contains no entries, `/var/log/auth.log` and `/var/log/syslog` are missing, and the login record files are empty.
3. **W4-F03 (Moderate): Vulnerable packages and 91 pending updates.** Lynis reported a warning for vulnerable packages, while `apt` showed 91 packages available for upgrade.

A fourth finding, **W4-F04 (Moderate)**, covers weak password settings and is discussed in the main report.

Highest priority. I consider W4-F01 the highest priority, followed closely by W4-F02. Without working audit and system logging, there is no reliable evidence of activities carried out on the system, which also makes it more difficult to verify whether other security controls are working as expected. The reasoning is explained further in Section 5.

2. Scope and Authorisation

All work for this lab was carried out inside the Ubuntu Practice Hub workspace assigned to me through the ICDFA E-Campus portal. I did not scan, test, or attempt to access any other system.

Item	What I recorded
Platform	ICDFA Ubuntu Practice Hub, a persistent Ubuntu 24.04 workspace with 1 CPU, 1.0 GB RAM, and 5.0 GB storage
Hostname and Username	<code>student000272</code> , logged in as <code>student</code>
Operating system	Ubuntu 24.04.4 LTS (Noble Numbat)
Kernel	Linux 7.0.0-1014-azure
First process (PID 1)	<code>sleep</code> . This means <code>systemd</code> is not running and suggests the workspace is operating as a container.
Privileges	The <code>student</code> account is in the <code>sudo</code> group, and <code>sudo id</code> returned <code>uid=0(root)</code>
Timestamp timezone	UTC. WAT is UTC+1, so 09:11 UTC corresponds to 10:11 WAT.

Boundaries I kept to. I did not edit `/etc/passwd` or `/etc/shadow`, delete or disable any logs, or upgrade the 91 pending packages. The only changes I made were installing Lynis and its dependency, `menu`, and creating `/etc/audit/rules.d/custom.rules`.

Auditd installation. The package history shows that `auditd` was installed on 1 October 2026 at 13:45:33 UTC, before this lab session. When I checked it today, `apt` reported that it was already the newest version, so I did not reinstall it.

Instructor contact. When I encountered the `auditd` issue, I was unable to reach my instructor before the deadline to request a different environment or an approved alternative. I therefore documented the failures as they occurred rather than trying to force the service or make changes outside the scope of the lab.

3. Methodology

I followed the lab manual module by module. For each activity, I ran the required command, captured the result, and recorded the evidence. When a command failed, I kept the exact error message instead of leaving out the failure. I then used the results to link each issue to the relevant control, owner, and follow-up action, which forms the governance part of the lab.

Activity	What I ran	What Happened
Module 1: auditd	<code>systemctl status auditd</code> , followed by <code>apt update</code> and <code>apt install auditd audispd-plugins</code> . I then tried <code>systemctl start/enable auditd</code> , <code>service auditd start/status</code> , and <code>auditctl -s</code> . I created custom rules, restarted the service, and checked <code>auditctl -l</code> , <code>auresearch -k passwd_changes</code> , and <code>aureport</code> .	<code>systemctl</code> could not work because <code>systemd</code> is not running. <code>auditd</code> was already installed. The start command failed, while enable completed through the older SysV script. The service was still not running. I created the rules file, but the restart failed because the rules could not be loaded, and no audit log was created.
Module 2: Logs	<code>journalctl</code> for all entries, SSH entries, today's entries, and errors. I also checked <code>/var/log/auth.log</code> and <code>/var/log/syslog</code> with <code>grep</code> , and ran <code>last</code> and <code>lastb</code> . I checked the APT history log as an alternative source.	The journal had no entries. <code>/var/log/auth.log</code> and <code>/var/log/syslog</code> do not exist, and the login record files were empty. The APT history log was the only log source that contained actual activity.
Module 3: Lynis	Installed Lynis with <code>apt install lynis</code> , checked <code>lynis --version</code> , ran <code>sudo lynis audit system</code> , and checked <code>/var/log/lynis-report.dat</code> .	Lynis 3.0.9 was installed from the Ubuntu repository. The audit returned a hardening index of 62, based on 232 tests, with 3 warnings and 43 suggestions. I also confirmed the score and warnings in the report file.
Governance mapping	Control-monitoring table, finding records, and SIEM mapping.	Completed in Sections 4 to 7 of this report.

What I did not do. I also recorded the activities I left out so the scope of the testing is clear.

- I only ran `ausearch` for the `passwd_changes` key. I did not run it for `program_execution` or `auth_failures`, and I did not run `aureport --failed` or `aureport --login`. Since no audit log existed, these checks would not have produced usable results.
- I did not open `/etc/passwd` in an editor and exit, or run the `ls /tmp` test. Since `auditd` was not running, these actions would not have generated audit records, and I did not want to open a sensitive file without a useful testing purpose.
- I did not add the `auth.log` audit rule because the manual specifies this only where `/var/log/auth.log` exists, and that file is not present on this system.
- I created the rules file using `tee` instead of `nano`. This created the same file and did not change the purpose of the step.
- I did not carry out the optional Activity 3.3 on safe hardening and retesting. No hardening changes were made.

4. Findings

4.1 auditd Configuration and Events (Evidence Bundle 1)

Service status. The first command in the manual, `systemctl status auditd`, returned: **"System has not been booted with systemd as init system (PID 1). Can't operate."** This error shows that `systemd` is not running, but it does not tell me whether `auditd` is installed. I therefore checked the package separately, and `auditd` and `audispd-plugins` were already installed at version `1:3.1.2-2.1ubuntu0.1`. The `which` command also confirmed that `auditd`, `auditctl`, `auresearch`, and `aureport` are available in `/usr/sbin`.

I then tried to start the service, but it failed with the same `systemd` error. The `enable` command completed without an error because it used the older SysV script. However, enabling the service does not mean that it is currently running. The process check showed no `auditd` process, and `service auditd status` confirmed that **"auditd is not running."** The attempt to start it with `service auditd start` also returned `[fail]`.

Rules. I created `/etc/audit/rules.d/custom.rules` containing four rules: watches for changes to `/etc/passwd` and `/etc/shadow`, using the keys `passwd_changes` and `shadow_changes`, and two `execve` rules for 64-bit and 32-bit programs using the `program_execution` key. The file was created successfully. However, when I tried to restart `auditd`, it failed again. Running `auditctl -l` returned **"Error sending rule list data request (Operation not permitted)."**

I was using `sudo` and had root privileges, so the result was not caused by a lack of permission within the workspace. The kernel refused the request. This is consistent with the workspace appearing to run in a container, although I could not confirm that this was the direct cause from inside the environment.

Queries. When I ran `auresearch -k passwd_changes` and `aureport`, both returned **"Error opening /var/log/audit/audit.log (No such file or directory)."** The `/var/log/audit` directory exists, but it is empty. Lynis also identified the same issue, showing **"Checking auditd [NOT FOUND]"** and recommending that `auditd` be enabled to collect audit information under `ACCT-9628`.

Interpretation. The purpose of the audit check is to use the records to establish **who performed an action, what they did, and when it happened**. In this case, there is no audit record to examine because `auditd` did not run and `/var/log/audit/audit.log` was never created. I therefore cannot use `auditd` evidence to identify who changed a sensitive file or executed a program.

The APT history provides some activity information, such as when packages were installed, but it is not a replacement for audit records. From a governance perspective, this shows why checking that a control is installed is not enough. `auditd` is present, but it is not producing

evidence that can be used to monitor or verify activity. The control therefore exists in configuration, but it is not providing the assurance it is supposed to provide.

4.2 Linux Log Analysis (Evidence Bundle 2)

Journal. I ran four `journalctl` checks: all entries, the SSH unit, entries from today, and entries with error priority. All four returned **"No journal files were found."** and **"-- No entries --"**. This means there was no journal data available for me to review.

Authentication and privilege use. `/var/log/auth.log` does not exist, so I could not use it to check failed passwords or `sudo` activity. I also checked the available login records. `btmpt`, `wtmp`, `lastlog`, and `faillog` were all 0 bytes, and both `last` and `lastb` returned no records.

This confirms that there are no recorded authentication events in these sources. However, an empty record does not prove that nobody attempted to log in. It could also mean that the attempts were not being captured. Lynis reported **"Logging failed login attempts [ENABLED]"**, which suggests the facility is enabled, but there are no records available to review.

Errors and warnings. `/var/log/syslog` is also missing, and there are no journal entries to provide another source for system errors and warnings. Lynis reported a related warning, **LOGG-2138**, stating that `klogd` is not running and kernel messages could be lost. It also gave suggestion **LOGG-2130**, which recommends checking that a syslog daemon is running.

Equivalent source used. The only log containing actual activity was `/var/log/apt/history.log`. I used this as an equivalent source for package-related privileged activity because installing packages requires `sudo`. It records the installation of `auditd` on **1 October 2026 at 13:45:33 UTC** and Lynis on **2 October 2026 at 09:07:02 UTC**. However, these entries do not identify the user who ran the commands, so I have not assumed who performed them.

Event / Condition	Evidence source	Time (UTC)	User / Service	Security significance	Recommended action
No journal entries exist	<code>journalctl</code> four queries	2 Oct 2026	systemd journal	System events cannot be reviewed, leaving no evidence for investigation	Enable a logging service and confirm that entries are being recorded
<code>auth.log</code> and <code>syslog</code> are missing	File checks and <code>grep</code>	2 Oct 2026	rsyslog / authentication logging	There is no available source for failed logins, <code>sudo</code> activity, errors, or warnings	Enable authentication and system logging
No failed logins recorded	<code>lastb</code> , <code>btmpt</code> , and <code>wtmp</code>	2 Oct 2026	Login records	The empty records do not prove that no attempts occurred because logging may not be collecting them	Enable login logging and retest
<code>auditd</code> installed	APT history and <code>dpkg</code> log	1 Oct 2026 13:45:33	<code>apt</code> with <code>sudo</code>	Confirms the package was installed, but does not show who installed it or that <code>auditd</code> ever ran	Forward APT and <code>sudo</code> activity to central logging
Lynis installed	APT history	2 Oct 2026 09:07:02	<code>apt</code> with <code>sudo</code>	Provides the only recorded privileged package activity for the current day	Forward APT and <code>sudo</code> activity to central logging
Security testing tools present in the image (<code>nmap</code> , <code>hydra</code> , <code>sqlmap</code> , <code>john</code> , and others)	APT history	16 Aug 2026	<code>apt</code>	These tools are normal for a training image, but their presence on a real system would require an authorisation review	None for this lab; noted as context only

Overall interpretation. The main issue is not simply that some log files are missing. The bigger problem is that there is no reliable logging source that allows activity on the system to be reconstructed. The APT history gives limited information about package changes, but it does not provide the user, command context, or broader system activity that would normally come from authentication, system, and audit logs.

4.3 Lynis Security Assessment (Evidence Bundle 3)

I installed Lynis 3.0.9 using `sudo apt install lynis` from the Ubuntu Noble repository on 2 October 2026 at 09:07 UTC. I then ran `sudo lynis audit system`. The scan finished at 09:11 UTC (10:11 WAT), based on the file timestamps. The results showed a **hardening index of 62**, with **232 tests performed, 3 warnings, and 43 suggestions**. The report is saved in `/var/log/lynis-report.dat`, while the full scan log is in `/var/log/lynis.log`. Lynis detected the firewall component but did not find a malware scanner.

Things that looked fine. Lynis reported OK for password file consistency, accounts without passwords, permissions on `/etc/sudoers` and `/etc/sudoers.d`, hardened mount options for `/tmp`, `/run`, and `/dev/shm`, the absence of telnet and rsh packages, and an active host-based firewall.

Findings I did not treat as weaknesses. Some results appear to be related to the way this workspace is set up rather than actual security weaknesses. These include the missing `/boot/vmlinuz` (KRNL-5788), boot loader and UEFI checks, systemd service suggestions (BOOT-5180 and BOOT-5264), and the kernel package exception (PKGS-7410). I left these out because they would not be meaningful findings on a normal Linux server based on the environment I was given.

Lynis ID	Risk/control issue	Evidence	Likely owner	Recommended remediation	Priority
LOGG-2130, LOGG-2138	No logging daemon is running, and <code>klogd</code> is not running	LOGG-2138 warning. Syslog-NG, RSyslog, journal, and <code>klogd</code> were all reported as NOT FOUND	IT Operations	Install and start an approved logging service, then confirm that entries are being recorded in the journal and <code>/var/log</code>	High
ACCT-9628	<code>auditd</code> is not collecting audit information	Lynis reported "Checking auditd: NOT FOUND". My own attempt to start the service also failed	System Administrator	Run <code>auditd</code> on a full VM or approved host, load the required rules, and confirm them with <code>auditctl -l</code>	High

PKGS-7392	Vulnerable or outdated packages are present	Lynis reported PKGS-7392, and apt showed 91 packages available for upgrade	IT Operations / Patching	Review the pending updates and apply required security updates through the normal change-control process	Moderate
AUTH-9286, AUTH-9262	Password ageing is disabled and no password-strength module is configured	Minimum and maximum password ages are disabled. Lynis also returned suggestions AUTH-9286 and AUTH-9262	System Administrator / IAM Owner	Configure password ageing in /etc/login.defs and add an appropriate PAM password-strength module in line with the organisation's password policy	Moderate
FINT-4350	No file integrity monitoring tool is present	Lynis reported "Checking presence integrity tool: NOT FOUND"	Security Operations	Deploy an appropriate integrity monitoring tool for critical system files	Moderate
DEB-0880	fail2ban is not installed	Lynis returned suggestion DEB-0880. It also found no running SSH daemon	Security Operations	Consider fail2ban if SSH or another remote login service is exposed	Low
NETW-2705	Fewer than two responsive nameservers	Lynis reported NETW-2705, with only 127.0.0.11 listed	Network Team	Add a backup nameserver if the environment allows it	Low

Lynis is useful for identifying possible weaknesses, but its suggestions are not automatically findings that need the same level of attention. I therefore considered the actual evidence and the likely security impact when assigning the priorities above, rather than simply grouping them by the number of Lynis recommendations.

4.4 Finding Records

The following finding records use the Appendix A template from the lab manual.

W4-F01: Audit Logging Does Not Work

Field	Entry
Finding ID	W4-F01
Date/time observed	2 Oct 2026 (UTC), based on my terminal output
Evidence Source	<code>auditd</code> service commands, <code>auditctl</code> , <code>ausearch</code> , <code>aureport</code> , <code>/var/log/audit</code> , and Lynis ACCT-9628
Evidence summary	<code>auditd</code> is installed but not running. Attempts to start and restart it failed, <code>auditctl -l</code> returned " Operation not permitted ", <code>/var/log/audit</code> is empty, and <code>ausearch</code> and <code>aureport</code> could not open <code>audit.log</code> .
Security significance	File access and program execution are not being recorded, so unauthorised or risky activity cannot be detected or investigated after the event.
Control objective	Security-relevant activity on the system is recorded, retained, and available for review.
Control Owner	System Administrator, with Security Operations using the evidence for monitoring and review
Risk / priority	High
Recommended action	Run <code>auditd</code> on a full VM or other approved host, load the required rules from <code>rules.d</code> , and confirm that they remain active after a restart.
Retest evidence	Service status showing active, <code>auditctl -l</code> listing the four rules, and <code>ausearch</code> returning events for the <code>passwd_changes</code> and <code>program_execution</code> keys.

W4-F02: System and Authentication Logs Are Not Collected

Field	Entry
Finding ID	W4-F02
Date/time observed	2 Oct 2026 (UTC), based on my terminal output
Evidence Source	<code>journalctl</code> , <code>/var/log</code> listing, <code>lastb</code> , <code>btmpt</code> , <code>wtmp</code> , and Lynis LOGG-2130 and LOGG-2138
Evidence summary	The journal has no entries, <code>/var/log/auth.log</code> and <code>/var/log/syslog</code> do not exist, and <code>btmpt</code> , <code>wtmp</code> , <code>lastlog</code> , and <code>faillog</code> are 0 bytes. Lynis also found no running logging daemon.
Security significance	Failed logins, <code>sudo</code> use, service errors, and kernel messages are not being retained. An empty result does not necessarily mean nothing happened because the events may simply not be recorded.
Control objective	Authentication, privilege use, and system events are logged and retained for review.
Control Owner	IT Operations, with Security Operations reviewing the logs
Risk / priority	High
Recommended action	Enable an approved syslog or journal service, ensure authentication events are written to a log, and forward the logs to central monitoring.
Retest evidence	<code>journalctl</code> showing recent entries, <code>/var/log/auth.log</code> present with <code>sudo</code> and login events, and a test failed login appearing in the log.

W4-F03: Vulnerable Packages and Pending Updates

Field	Entry
Finding ID	W4-F03
Date/time observed	2 Oct 2026 (UTC), based on apt update and the Lynis scan
Evidence Source	apt update output and Lynis warning PKGS-7392
Evidence summary	apt reported 91 packages available for upgrade, and Lynis warned about vulnerable packages. I did not run an upgrade, so the evidence does not show which packages are affected or the severity of each issue.
Security significance	Known weaknesses in installed software may remain exploitable until the required updates are applied.
Control objective	Security updates are applied within the organisation's agreed timeframe.
Control Owner	IT Operations (patch management)
Risk / priority	Moderate. This should be raised if any affected package is critical or the system is exposed to the internet.
Recommended action	Review the list of upgradable packages and apply required security updates through the change-control process. Automatic security updates can also be considered, as suggested by Lynis PKGS-7420.
Retest evidence	apt showing no pending security updates and the PKGS-7392 warning no longer appearing in a new Lynis scan.

W4-F04: No Password Ageing or Strength Rules

Field	Entry
Finding ID	W4-F04
Date/time observed	2 Oct 2026 (UTC), based on the Lynis scan
Evidence Source	Lynis AUTH-9286, AUTH-9262, and AUTH-9230
Evidence summary	Minimum and maximum password ages are both shown as DISABLED , and Lynis recommends configuring a PAM module for password strength testing.
Security significance	Passwords may remain unchanged indefinitely and may not have sufficient strength requirements, which can increase exposure to guessing or password reuse attacks.
Control objective	Passwords follow the organisation's defined strength and ageing requirements.
Control Owner	System Administrator, with the Identity and Access Management owner
Risk / priority	Moderate
Recommended action	Configure the password ageing values in <code>/etc/login.defs</code> to match the organisation's password policy and add an appropriate PAM password-strength module.
Retest evidence	<code>/etc/login.defs</code> showing the new settings and the relevant AUTH tests passing in a new Lynis scan.

5. Control-Monitoring Table

The seven controls below are based on the evidence I collected during the lab. The KPI, KRI, and threshold values are my suggestions for this lab exercise. In a real organisation, these would be defined by the organisation's policies and risk requirements.

Control / Objective	Evidence	Owner	Observed status	KPI/KRI threshold	Risk / significance	Remediation	Retest / follow-up
1. System auditing: security events are recorded by auditd	service auditd start returned [fail]. auditctl -l: Operation not permitted. /var/log/audit empty. Lynis ACCT-9628	System Administrator	Not operating	KPI: auditd running on all in-scope servers. KRI: auditd stopped or silent for more than 24 hours	No evidence of file or program activity. Investigations are not possible	Run auditd on a supported host, load rules from rules.d, confirm with auditctl -l	Weekly status check until closed, then monthly. ausearch must return events for the custom keys
2. System log collection: journal, auth.log and syslog	journalctl: No journal files were found. auth.log and syslog missing. Lynis LOGG-2130 and LOGG-2138	IT Operations	Not operating	KPI: every in-scope host sends logs daily. KRI: no log events from a host for 24 hours	Failed logins, sudo use and errors are not kept	Enable a logging service and forward logs to central monitoring	Check for fresh journal entries and an auth log within 7 days of the fix, then monthly
3. Failed login monitoring	btmpt, wtmpt, lastlog at 0 bytes. lastb empty. Lynis shows failed login logging ENABLED	Security Operations	Cannot be confirmed (facility present, no records)	Alert: more than 5 failed logins for one account in 10 minutes. KRI: repeated failures across several systems	Empty records could hide real attempts	Fix logging (rows 1 and 2), then set the alert rule	Run a controlled failed login in the lab and confirm it appears in the log
4. Patch management: security updates applied on time	apt update: 91 packages can be upgraded. Lynis warning PKGS-7392	IT Operations	Gap found	KPI: critical and high security updates applied within 14 days. KRI: any critical update older than 14 days	Known vulnerabilities may stay open	Review the list, patch through change control, consider automatic security updates	Re-run apt update and Lynis after patching. Check monthly
5. Password policy: ageing and strength	Lynis: minimum and maximum password age DISABLED. AUTH-9286 and AUTH-9262	System Administrator and IAM owner	Not configured	KPI: all accounts covered by ageing and strength rules. KRI: any account with no ageing set	Weak or never-changed passwords raise the chance of account compromise	Set values in /etc/login.defs and add a PAM strength module	Check login.defs and re-run Lynis. Review quarterly

6. File integrity monitoring	Lynis: Checking presence integrity tool NOT FOUND. FINT-4350	Security Operations	Not in Place	KPI: integrity tool on all critical hosts. KRI: unexpected change to a monitored file	Changes to critical files could go unseen	Deploy an integrity tool for critical files	Confirm the tool reports a test change, then review monthly
7. Configuration hardening baseline	Lynis hardening index 62, 232 tests, 3 warnings, 43 suggestions	System Administrator and GRC	Moderate, with gaps	KPI: index at or above a target set by management (I suggest 80). KRI: a drop of 5 points or more between scans	A single score is a rough guide, but a falling score shows drift	Work through warnings first, then suggestions that fit the system's purpose	Re-run Lynis after each round of fixes and keep the scores in a trend sheet

Governance Escalation Questions

1. Which finding is the highest priority and why?

W4-F01, the audit logging failure, is the highest priority, with W4-F02, the lack of system and authentication logs, closely related. The two findings affect the ability to detect, investigate, and provide evidence of security events. Other issues, such as pending updates or weak password settings, can be identified through the available checks, but without working logging, it is difficult to know what is happening on the system or confirm whether other controls are working as expected.

2. Who should own remediation for each major finding?

W4-F01 should be owned by the System Administrator, with Security Operations reviewing the evidence. W4-F02 should be owned by IT Operations, with Security Operations responsible for reviewing the collected logs. W4-F03 should be handled by IT Operations through the patch management process, while W4-F04 should be owned by the System Administrator and the Identity and Access Management function. GRC should track all four findings until they are closed.

3. What threshold or condition should trigger escalation to the CISO, the Risk function or a management committee?

These thresholds are my suggested escalation points. An auditing or logging control that remains down on an in-scope server for more than 24 hours should be escalated to the CISO. A repeated failure during retesting or a critical update that remains unapplied for more than 14 days should be escalated to the Risk function. A High finding that remains open past its agreed due date, or a control failure that is found to have prevented the detection of a real incident, should be escalated to the management committee.

4. What evidence would show successful remediation?

For W4-F01, successful remediation would be shown by an active auditd service, `auditctl -l` displaying the required rules, and `aureport` returning events for the configured keys. For W4-F02, there should be recent `journalctl` entries, an `auth.log` containing sudo and login activity, and a controlled failed login test recorded in the logs. For W4-F03, there should be no pending security updates and the PKGS-7392 warning should no longer appear. For W4-F04, the updated `login.defs` settings should be in place and the related AUTH checks should pass in Lynis.

5. When and how should the control be retested?

Each finding should be retested within 7 days after remediation using the same commands used during the initial assessment. This makes it easier to compare the before and after evidence. After remediation, auditing and logging should be checked monthly, while Lynis should be run

after significant changes and at least once every quarter. The hardening index should also be recorded each time to track changes over time.

6. SIEM and Continuous Monitoring Mapping (Evidence Bundle 5)

I did not set up a SIEM because this part of the lab is conceptual. The monitoring flow I have in mind is: Linux host (auditd, journal, authentication logs and Lynis report) → log forwarder or agent → SIEM → alerts, dashboard and ticket. The table below shows how the evidence sources from this lab could fit into that process and the difference between a technical alert handled by the security team and a control issue that requires governance attention.

Evidence Source	What I saw in this lab	How it could reach a SIEM	Technical alert (SOC handles it)	Governance / control-assurance issue
auditd	Not running. No audit log	An agent or audit plugin sends events (auditd-plugins is already installed on this host)	A change to /etc/passwd or /etc/shadow by an unexpected user. An unusual program run by a service account	auditd stopped, or no audit events from a host for 24 hours
journalctl and syslog	No journal files. No syslog	Syslog forwarding or an agent reads the journal	Service crashes and kernel errors	A host that sends no logs at all, which the SIEM can detect as a silent source
Authentication log and login records	auth.log missing. bttmp, wttmp and lastb empty	An agent reads auth.log or the journal for the ssh and sudo units	Repeated failed passwords on one account. sudo used by an unexpected user	Failures spread across many hosts, or a server with no authentication logging
Lynis report	Index 62. 3 warnings. 43 suggestions	A scheduled scan, with /var/log/lynis-report.dat collected as structured data	A new critical warning appearing after a change	A falling index, or findings still open after their due date
apt and dpkg history	Installs of auditd and Lynis recorded with times	Collected as software change evidence	An unexpected package installed	A growing patch backlog, or unapproved software across systems

7. Remediation and Retest Plan

The target times below are my proposed timelines. In a real organisation, the final timelines would be agreed with the control owners and management.

Finding	Action	Owner	Proposed Target	Evidence of closure	Retest
W4-F01 (High)	Run auditd on a supported host, put the rules in rules.d, load them and confirm	System Administrator	Within 7 days	Status active, auditctl -l lists the rules, ausearch returns events	Repeat the Module 1 commands within 7 days, then monthly
W4-F02 (High)	Enable a logging service, enable authentication logging, forward to central monitoring	IT Operations	Within 7 days	Journal entries, auth.log present, test failed login is visible	Repeat the Module 2 commands within 7 days, then monthly
W4-F03 (Moderate)	Review and apply security updates through change control	IT Operations	Within 14 days	No pending security updates, PKGS-7392 gone	Re-run apt update and Lynis, then monthly
W4-F04 (Moderate)	Set password ageing and add a strength module	System Administrator and IAM owner	Within 30 days	New login.defs values, AUTH checks pass	Re-run Lynis, then quarterly

8. Conclusion

Before this lab, I thought monitoring was mainly about installing the right tools. From the practical work, I learned that having a tool installed does not mean the control is actually working. auditd was installed and I created the required rules, but it did not produce any audit records. The usual logs for login and sudo activity were also unavailable. The Lynis scan showed a hardening score of 62 out of 100, with gaps in patching, password settings and file integrity monitoring.

The workspace also appears to have limited some of the tests because it seems to be a container-based environment. I followed the lab manual, documented the failures as they occurred and did not create evidence that I could not verify. This was an important governance lesson for me because security decisions should be based on actual evidence. If a control is not working or cannot be verified, that needs to be clearly recorded.

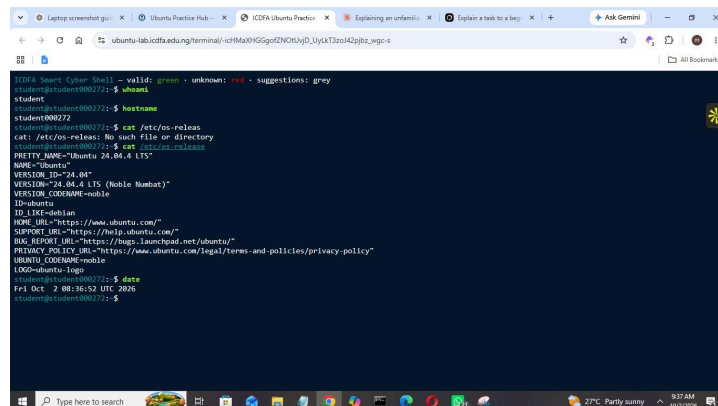
If this were a real system, I would first address W4-F01 and W4-F02 by restoring auditing and logging. These controls provide the evidence needed to monitor and investigate other security activities. I would then address the patching and password issues and retest each control using the same commands, so the results before and after remediation can be compared.

Appendix: Command Output and Screenshots

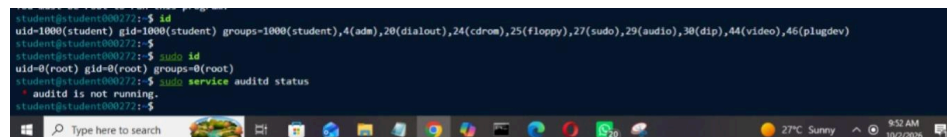
The output below is copied from my own terminal session on 2 October 2026 (times in UTC). Where an output was long, I shortened it and marked it.

A1. Environment

```
student@student000272:~$ ps -p 1 -o comm=
sleep
student@student000272:~$ uname -a
Linux student000272 7.0.0-1014-azure #14~24.04.1-Ubuntu SMP PREEMPT Fri Aug 14 17:35:51 UTC 2026 x86_64 x86_64
x86_64 GNU/Linux
student@student000272:~$ whoami
student
student@student000272:~$ hostname
student000272
student@student000272:~$ date
Fri Oct 2 08:57:44 UTC 2026
student@student000272:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
student@student000272:~$ id
uid=1000(student) gid=1000(student)
groups=1000(student),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev)
student@student000272:~$ sudo id
uid=0(root) gid=0(root) groups=0(root)
student@student000272:~$ uptime
09:40:22 up 4:40, 0 user, load average: 0.11, 0.10, 0.09
```

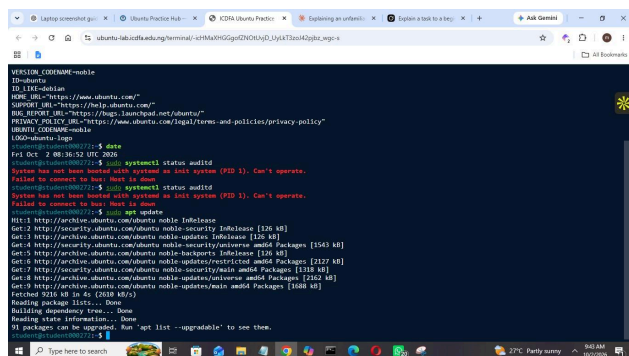
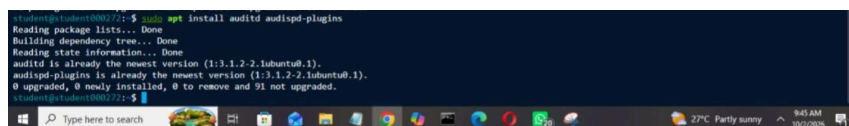


Terminal showing User, hostname, Ubuntu 24.04.4, date in UTC



A2. auditd service checks (Module 1)

```
student@student000272:~$ sudo systemctl status auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
student@student000272:~$ sudo apt install auditd audispd-plugins
auditd is already the newest version (1:3.1.2-2.1ubuntu0.1).
audispd-plugins is already the newest version (1:3.1.2-2.1ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 91 not upgraded.
student@student000272:~$ which auditd auditctl ausearch aureport
/usr/sbin/auditd
/usr/sbin/auditctl
/usr/sbin/ausearch
/usr/sbin/aureport
student@student000272:~$ sudo systemctl start auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
student@student000272:~$ sudo systemctl enable auditd
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
student@student000272:~$ ps aux | grep '[a]uditd'
(no output)
student@student000272:~$ sudo service auditd status
* auditd is not running.
student@student000272:~$ sudo service auditd start
* Starting audit daemon auditd                                [fail]
student@student000272:~$ sudo auditctl -s
You must be root to run this program.
student@student000272:~$ sudo ls -l /var/log/audit/
total 0
```

A screenshot of a terminal window on a Linux system. The terminal shows the output of several commands. First, it displays system information including the kernel version (5.15.0-56-generic) and the systemd version (249.3-2ubuntu3.1). Then, it shows the output of the 'sudo apt update' command, which lists several packages to be updated, including 'auditd', 'audispd-plugins', and 'libaudit1'. The terminal also shows the output of 'sudo apt install auditd audispd-plugins', which indicates that the packages are already installed and up-to-date. The terminal window has a dark background and a light-colored text. The window title is 'Ubuntu Practice Hub'. The window also shows a search bar at the top and a taskbar at the bottom with various application icons.A screenshot of a terminal window showing the installation of auditd and audispd-plugins. The terminal displays the output of the command 'sudo apt install auditd audispd-plugins'. The output shows that the packages are already installed and up-to-date. The terminal window has a dark background and a light-colored text. The window title is 'Ubuntu Practice Hub'. The window also shows a search bar at the top and a taskbar at the bottom with various application icons.

```
Laptop screenshot gu... Ubuntu Practice Hub... KORA Ubuntu Practi... Explaining an unfami... Explain a task to a be... Ask Gemini
ubuntu-lab.icfda.edung/terminal/-icHMaXHG6gzNCRLuYD_UyLkT3zo42pbhz_wgc-s
Get:4 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1543 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2127 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1318 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2162 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1688 kB]
Fetched 9216 kB in 4s (2618 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
91 packages can be upgraded. Run 'apt list --upgradable' to see them.
student@student000277:~$ sudo apt install auditd auditd-plugins
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
auditd is already the newest version (1:3.1.2-2.1ubuntu1).
auditd-plugins is already the newest version (1:3.1.2-2.1ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 91 not upgraded.
student@student000277:~$ sudo auditd auditd modctl ausersearch auserport
/usr/sbin/auditd
/usr/sbin/auditctl
/usr/sbin/aureport
student@student000277:~$ sudo systemctl start auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
student@student000277:~$ sudo systemctl enable auditd
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
student@student000277:~$ ps aux | grep '[a]uditd'
student@student000277:~$ sudo auditctl -s
You must be root to run this program.
student@student000277:~$
```

```
student@student000277:~$ sudo service auditd start
* Starting audit daemon auditd                                     [fail]
student@student000277:~$ sudo service auditd status
auditd is not running.
student@student000277:~$ sudo auditctl -s
You must be root to run this program.
student@student000277:~$
```

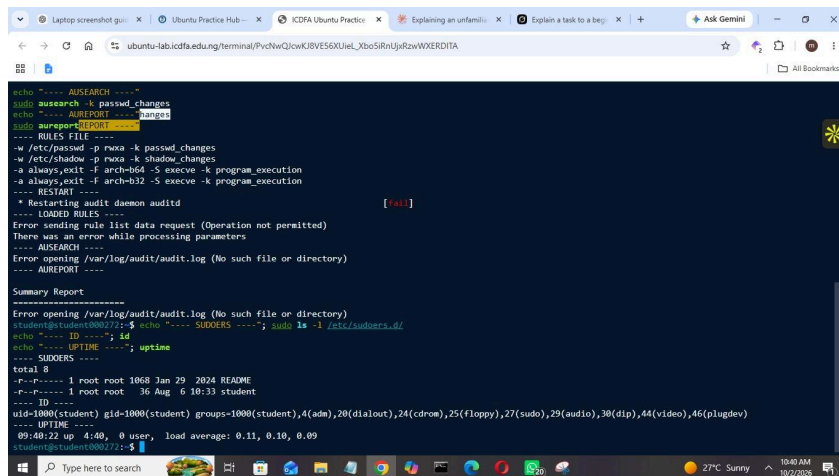
A3. Audit rules and queries (Module 1)

```
student@student000272:~$ sudo cat /etc/audit/rules.d/custom.rules
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
student@student000272:~$ sudo service auditd restart
* Restarting audit daemon auditd [fail]
student@student000272:~$ sudo auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
student@student000272:~$ sudo ausearch -k passwd_changes
Error opening /var/log/audit/audit.log (No such file or directory)
student@student000272:~$ sudo aureport
```

Summary Report

=====

Error opening /var/log/audit/audit.log (No such file or directory)



The screenshot shows a terminal window with the following content:

```
echo "---- AUSEARCH ----"
sudo ausearch -k passwd_changes
echo "---- AUREPORT ----"
sudo aureport -k passwd_changes
---- RULES FILE ----
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
---- RESTART ----
* Restarting audit daemon auditd [fail]
---- LOADED RULES ----
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
---- AUSEARCH ----
Error opening /var/log/audit/audit.log (No such file or directory)
---- AUREPORT ----

Summary Report
=====
Error opening /var/log/audit/audit.log (No such file or directory)
student@student000272:~$ echo "---- SUDOERS ----"; sudo ls -l /etc/sudoers.d/
echo "---- ID ----"; id
echo "---- UPTIME ----"; uptime
---- SUDOERS ----
total 8
-rw-r--r-- 1 root root 1068 Jan 29 2024 README
-rw-r--r-- 1 root root 36 Aug 6 10:33 student
---- ID ----
uid=1000(student) gid=1000(student) groups=1000(student),4(ade),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev)
---- UPTIME ----
09:48:22 up 4:48, 0 user, load average: 0.11, 0.10, 0.09
student@student000272:~$
```

A4. Log review (Module 2)

```
student@student000272:~$ sudo journalctl --no-pager | head -n 20
No journal files were found.
-- No entries --
(the queries for -u ssh, --since "today" and -p err returned the same two lines)
student@student000272:~$ ls -l /var/log/auth.log /var/log/syslog
ls: cannot access '/var/log/auth.log': No such file or directory
ls: cannot access '/var/log/syslog': No such file or directory
student@student000272:~$ ls -l /var/log
-rw-rw---- 1 root utmp      0 Jul 30 14:19 btmp
-rw-r--r-- 1 root root      0 Jul 30 14:19 faillog
-rw-rw-r-- 1 root utmp      0 Jul 30 14:19 lastlog
-rw-rw-r-- 1 root utmp      0 Jul 30 14:19 wtmp
(other entries shortened: alternatives.log, apt, bootstrap.log, dpkg.log, audit, journal, private, samba)
student@student000272:~$ last | head -n 10
wtmp begins Thu Aug  6 04:28:58 2026
student@student000272:~$ sudo lastb | head -n 10
btmp begins Thu Aug  6 04:28:58 2026
student@student000272:~$ grep -i auditd /var/log/dpkg.log | head -n 10
2026-10-01 13:45:34 install auditd:amd64 <none> 1:3.1.2-2.1ubuntu0.1
2026-10-01 13:45:34 status half-installed auditd:amd64 1:3.1.2-2.1ubuntu0.1
2026-10-01 13:45:34 status unpacked auditd:amd64 1:3.1.2-2.1ubuntu0.1
2026-10-01 13:45:34 configure auditd:amd64 1:3.1.2-2.1ubuntu0.1 <none>
2026-10-01 13:45:34 status unpacked auditd:amd64 1:3.1.2-2.1ubuntu0.1
2026-10-01 13:45:34 status half-configured auditd:amd64 1:3.1.2-2.1ubuntu0.1
2026-10-01 13:45:34 status installed auditd:amd64 1:3.1.2-2.1ubuntu0.1
student@student000272:~$ sudo grep -E "Start-Date|Commandline" /var/log/apt/history.log | tail -n 12
(shortened to the relevant entries)
Start-Date: 2026-10-01 13:45:33
Commandline: apt install auditd audispd-plugins
Start-Date: 2026-10-02 09:07:02
Commandline: apt install lynis
(earlier entries dated 16 Aug 2026 are the workspace image setup and include tools such as nmap, hydra, sqlmap and john)
student@student000272:~$ sudo ls -l /etc/sudoers.d/
-r--r----- 1 root root 1068 Jan 29  2024 README
-r--r----- 1 root root  36 Aug  6 10:33 student
```


A5. Lynis assessment (Module 3)

```
student@student000272:~$ apt-cache policy lynis
lynis:
  Installed: (none)
  Candidate: 3.0.9-1
student@student000272:~$ sudo apt install lynis
The following NEW packages will be installed:
  lynis menu
Setting up lynis (3.0.9-1) ...
student@student000272:~$ lynis --version
3.0.9
student@student000272:~$ sudo lynis audit system
(full scan output shortened to the results section)
Warnings (3):
! Found one or more vulnerable packages. [PKGS-7392]
! Couldn't find 2 responsive nameservers [NETW-2705]
! klogd is not running, which could lead to missing kernel messages in log files [LOGG-2138]

Suggestions (43):
(examples)
* Enable auditd to collect audit information [ACCT-9628]
* Check if any syslog daemon is running and correctly configured. [LOGG-2130]
* Configure minimum password age in /etc/login.defs [AUTH-9286]
* Install a PAM module for password strength testing like pam_cracklib or pam_passwdqc [AUTH-9262]
* Install a file integrity tool to monitor changes to critical and sensitive files [FINT-4350]
* Install fail2ban to automatically ban hosts that commit multiple authentication errors. [DEB-0880]

Lynis security scan details:
Hardening index : 62 [##### ]
Tests performed : 232
Plugins enabled : 1
Components:
- Firewall           [V]
- Malware scanner    [X]
Files:
- Test and debug information : /var/log/lynis.log
- Report data               : /var/log/lynis-report.dat
student@student000272:~$ sudo grep hardening_index /var/log/lynis-report.dat
hardening_index=62
student@student000272:~$ sudo grep -E "^warning\[|\]" /var/log/lynis-report.dat
warning[]=PKGS-7392|Found one or more vulnerable packages.|-|
warning[]=NETW-2705|Couldn't find 2 responsive nameservers|-|
warning[]=LOGG-2138|klogd is not running, which could lead to missing kernel messages in log files|-|
student@student000272:~$ ls -l /var/log/lynis.log /var/log/lynis-report.dat
-rw-r----- 1 root root 40245 Oct  2 09:11 /var/log/lynis-report.dat
-rw-r----- 1 root root 438898 Oct  2 09:11 /var/log/lynis.log
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  menu
Suggested packages:
  apt-listbugs debconf debsums tripwire sunmail aide fail2ban menu-110n gksu
The following NEW packages will be installed:
  lynix menu
0 upgraded, 2 newly installed, 0 to remove and 91 not upgraded.
Need to get 682 kB of archives.
After this operation, 3502 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 lynix all 3.0.9-1 [226 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 menu amd64 2.1.50 [377 kB]
Fetched 602 kB in 2s (312 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package lynix.
(Reading database ... 41145 files and directories currently installed.)
Preparing to unpack .../archives/lynix_3.0.9-1_all.deb ...
Unpacking lynix (3.0.9-1) ...
Selecting previously unselected package menu.
Preparing to unpack .../archives/menu_2.1.50_amd64.deb ...
Unpacking menu (2.1.50) ...
Setting up lynix (3.0.9-1) ...
Created symlink /etc/systemd/system/timers.target.wants/lynix.timer → /usr/lib/systemd/system/lynix.timer.
Setting up menu (2.1.50) ...
Processing triggers for man-db (2.12.8-4build2) ...
Processing triggers for menu (2.1.50) ...
student@tudent0802771:~$ lynix --version
3.0.9
```

```
2026-10-01 13:45:34 configure auditd:amd64 1:3.1.2-2.1ubuntu0.1 cosmos
2026-10-01 13:45:34 status unpacked auditd:amd64 1:3.1.2-2.1ubuntu0.1
2026-10-01 13:45:34 status half-configured auditd:amd64 1:3.1.2-2.1ubuntu0.1
2026-10-01 13:45:34 status installed auditd:amd64 1:3.1.2-2.1ubuntu0.1
student@tudent0802771:~$ sudo lastls | head -n 10

btmp begins Thu Aug 6 04:28:58 2026
student@tudent0802771:~$ sudo lynix audit system

[ Lynix 3.0.9 ]

#####
Lynix comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.

2007-2021, CISOfy - https://cisofy.com/lynix/
Enterprise support available (Compliance, plugins, interface and tools)
#####

[+] Initializing program
-----
- Detecting OS... [ DONE ]
- Checking profiles... [ DONE ]
-----
Program version: 3.0.9
Operating system: Linux
Operating system name: Ubuntu
Operating system version: 24.04
Kernel version: 7.0.0
Hardware platform: x86_64
```

```
Checking password hashing rounds [ DISABLED ]
Query system users (non daemons) [ DONE ]
- NIS+ authentication support [ NOT ENABLED ]
- NIS authentication support [ NOT ENABLED ]
- Sudoers files(s) [ FOUND ]
- Permissions for directory: /etc/sudoers.d [ WARNING ]
- Permissions for: /etc/sudoers [ OK ]
- Permissions for: /etc/sudoers.d/student [ OK ]
- Permissions for: /etc/sudoers.d/README [ OK ]
PAM password strength tools [ SUGGESTION ]
- PAM configuration files (pam.conf) [ FOUND ]
- PAM configuration files (pam.d) [ FOUND ]
- PAM modules [ FOUND ]
LDAP module in PAM [ NOT FOUND ]
Accounts without expire date [ FOUND ]
Accounts without password [ OK ]
Locked accounts [ FOUND ]
Checking user password aging (minimum) [ DISABLED ]
User password aging (maximum) [ DISABLED ]
Checking expired passwords [ OK ]
Checking Linux single user mode authentication [ OK ]
Determining default umask [ NOT FOUND ]
- umask (/etc/profile) [ SUGGESTION ]
- umask (/etc/login.defs) [ SUGGESTION ]
LDAP authentication support [ NOT ENABLED ]
Logging failed login attempts [ ENABLED ]

[+] Shells
-----
- Checking shells from /etc/shells [ FOUND ]
Result: found 11 shells (valid shells: 11).
- Session timeout settings/tools [ NONE ]
Checking default umask values [ FOUND ]
```

```
ubuntu-lab.icdf.aadung/terminal/~chMaxHGgofZNOuJD_UyKAT3zi42pht_wgc-s

[+] Storage
- Checking usb-storage driver (modprobe config) [ NOT DISABLED ]
- Checking USB devices authorization [ DISABLED ]
- Checking USBGuard [ NOT FOUND ]

[+] NFS
- Checking firewire ohci driver (modprobe config) [ NOT DISABLED ]
- Check running NFS daemon [ NOT FOUND ]

[+] Name services
- Checking search domains [ FOUND ]
- Checking /etc/resolv.conf options [ FOUND ]
- Searching DNS domain name [ UNKNOWN ]
- Checking /etc/hosts
  - Duplicate entries in hosts file [ NONE ]
  - Presence of configured hostname in /etc/hosts [ FOUND ]
  - Hostname mapped to localhost [ NOT FOUND ]
  - Localhost mapping to IP address [ OK ]

[+] Ports and packages
- Searching package managers [ FOUND ]
- Searching deb package manager [ FOUND ]
- Querying package manager [ NONE ]
- Querying rpm packages [ OK ]
- Checking security repository in sources.list.d directory [ OK ]
- Checking apt package database [ OK ]
- Checking vulnerable packages [ WARNING ]
```

```
ubuntu-lab.icdf.aadung/terminal/~chMaxHGgofZNOuJD_UyKAT3zi42pht_wgc-s

[+] SNMP Support
- Checking running SNMP daemon [ NOT FOUND ]

[+] Databases
- No database engines found

[+] LDAP Service
- Checking OpenLDAP instance [ NOT FOUND ]

[+] PHP
- Checking PHP [ NOT FOUND ]

[+] Squid Support
- Checking running Squid daemon [ NOT FOUND ]

[+] Logging and files
- Checking for a running log daemon [ WARNING ]
- Checking Syslog-MG status [ NOT FOUND ]
- Checking syslogd journal status [ NOT FOUND ]
- Checking Metalog status [ NOT FOUND ]
- Checking Rsyslog status [ NOT FOUND ]
- Checking rsyslog daemon status [ NOT FOUND ]
- Checking klogd [ NOT FOUND ]
- Checking minilogd instances [ NOT FOUND ]
- Checking logrotate presence [ OK ]
- Checking log directories (static list) [ DONE ]
```

```
ubuntu-lab.icdf.aadung/terminal/~chMaxHGgofZNOuJD_UyKAT3zi42pht_wgc-s

[+] Insecure services
- Installed inetd package [ NOT FOUND ]
- Installed xinetd package [ OK ]
- xinetd status [ OK ]
- Installed rsh client package [ OK ]
- Installed rsh server package [ OK ]
- Installed telnet client package [ OK ]
- Installed telnet server package [ NOT FOUND ]
- Checking RLS client installation [ OK ]
- Checking RLS server installation [ OK ]
- Checking TFTP client installation [ OK ]
- Checking TFTP server installation [ OK ]

[+] Banners and Identification
- /etc/issue [ FOUND ]
- /etc/issue contents [ WEAK ]
- /etc/issue.net [ FOUND ]
- /etc/issue.net contents [ WEAK ]

[+] Scheduled tasks
- Checking cronstab and cronjob files [ DONE ]

[+] Accounting
- Checking accounting information [ NOT FOUND ]
- Checking sysstat accounting data [ NOT FOUND ]
- Checking auditd [ NOT FOUND ]

[+] Time and Synchronization
```

```
Laptop screenshot gu... Ubuntu Practice Hub... ICDA Ubuntu Practice... Explaining an unf... Explain a task to a be... Ask Gemini
ubuntu-lab.icda.edu.ng/terminal/-iCHMAxHGogfZNOHjD_UyXkT3co42pht_wgc-s
[+] Time and Synchronization
-----
[+] Cryptography
-----
- Checking for expired SSL certificates [0/122] [ NONE ]
- Kernel entropy is sufficient [ YES ]
- HW RNG & rngd [ NO ]
- SW prng [ NO ]
- HW veritable not found [ none ]
[+] Virtualization
-----
[+] Containers
-----
[+] Security Frameworks
-----
- Checking presence AppArmor [ NOT FOUND ]
- Checking presence SELinux [ NOT FOUND ]
- Checking presence TKNVD Linux [ NOT FOUND ]
- Checking presence grsecurity [ NOT FOUND ]
- Checking for implemented MAC framework [ NONE ]
[+] Software: file integrity
-----
- Checking file integrity tools [ NOT FOUND ]
- Checking presence integrity tool [ NOT FOUND ]
[+] Software: System tooling
-----
```

```
Laptop screenshot gu... Ubuntu Practice Hub... ICDA Ubuntu Practice... Explaining an unf... Explain a task to a be... Ask Gemini
ubuntu-lab.icda.edu.ng/terminal/-iCHMAxHGogfZNOHjD_UyXkT3co42pht_wgc-s
- Non-native binary formats [ NOT FOUND ]
[+] Custom tests
-----
- Running custom tests... [ NONE ]
[+] Plugins (phase 2)
-----
-----
[ Lynda 3.0.9 Results ]:
-----
Warnings (3):
-----
- Found one or more vulnerable packages, [PKGS-7392]
  https://cisofy.com/lynis/controls/PKGS-7392/
- Couldn't find 2 responsive nameservers [NLSW-2705]
  https://cisofy.com/lynis/controls/NLSW-2705/
- klogd is not running, which could lead to missing kernel messages in log files [LOG-2138]
  https://cisofy.com/lynis/controls/LOG-2138/
Suggestions (4):
-----
- This release is more than 4 months old. Check the website or Github to see if there is an update available. [LYN25]
  https://cisofy.com/lynis/controls/LYN25/
- Install lilo-pam-tapir to set $TDP and $TDPDIR for PAM sessions [DEB-0288]
  https://cisofy.com/lynis/controls/DEB-0288/
- Install apt-listbugs to display a list of critical bugs prior to each APT installation. [DEB-0810]
```

```
Laptop screenshot gu... Ubuntu Practice Hub... ICDA Ubuntu Practice... Explaining an unf... Explain a task to a be... Ask Gemini
ubuntu-lab.icda.edu.ng/terminal/F-3fCpH_RCBYvUUL0bz79d5GyQZvRQwKq03q99dM
lynda Smart Cyber Shell - valid: green - unknown: red - suggestions: grey
student@student0002721:~$ uname
student
student@student0002721:~$ lynda --version
3.0.9
student@student0002721:~$ ls -l /var/log/lynis.log /var/log/lynis-report.dat
-rw-r--r-- 1 root root 40385 Oct 2 09:11 /var/log/lynis-report.dat
-rw-r--r-- 1 root root 438898 Oct 2 09:11 /var/log/lynis.log
student@student0002721:~$ grep -e warning /var/log/lynis-report.dat
warning:[PKGS-7392]found one or more vulnerable packages.[1-]
warning:[NLSW-2705]couldn't find 2 responsive nameservers.[1-]
warning:[LOG-2138]klogd is not running, which could lead to missing kernel messages in log files[1-]
student@student0002721:~$ grep -i auditd /var/log/lynis.log | head -n 10
2026-10-01 13:45:34 install auditd:auditd cronos 1:3.1.2-2.1ubuntu1
2026-10-01 13:45:34 status half-installed auditd:auditd 1:3.1.2-2.1ubuntu1
2026-10-01 13:45:34 status unpacked auditd:auditd 1:3.1.2-2.1ubuntu1
2026-10-01 13:45:34 configure auditd:auditd 1:3.1.2-2.1ubuntu1 <cron>
2026-10-01 13:45:34 status unpacked auditd:auditd 1:3.1.2-2.1ubuntu1
2026-10-01 13:45:34 status half-configure auditd:auditd 1:3.1.2-2.1ubuntu1
2026-10-01 13:45:34 status installed auditd:auditd 1:3.1.2-2.1ubuntu1
student@student0002721:~$
session ended. You may close this tab.
```